

ASCII MASTER FLOW – PANEL 1/12: Seismic risk grammar

HAZARD -> ground shaking and secondary effects

EXPOSURE -> people buildings infrastructure lifelines

VULNERABILITY -> site + design + construction + maintenance

CAPACITY -> preparedness response and recovery

ASCII MASTER FLOW – PANEL 2/12: Magnitude-intensity firewall

MAGNITUDE -> event size / energy measure

INTENSITY -> observed effects at a location

ONE EVENT -> varying local intensities

TRAP -> scales are not interchangeable

ASCII MASTER FLOW – PANEL 3/12: Zoning-to-site ladder

SEISMIC ZONE -> broad expected hazard
MICROZONATION -> finer local variation
SITE ASSESSMENT -> project-specific conditions
NONE OF THESE -> predicts the next event

ASCII MASTER FLOW – PANEL 4/12: Site-effect map

GROUND / SLOPE / SOIL CONDITION

-> modifies shaking or secondary risk

BUILT FORM + ACCESS

-> modifies exposure response and loss

ASCII MASTER FLOW – PANEL 5/12: Vulnerability chain

IRREGULAR FORM | WEAK CONNECTION | POOR WORKMANSHIP
UNAUTHORISED ALTERATION | DEGRADED MATERIAL
FAILED APPROVAL / INSPECTION / MAINTENANCE
RESULT -> collapse or service disruption risk

ASCII MASTER FLOW – PANEL 6/12: Mitigation matrix

STRUCTURAL -> safe new build + assessed strengthening

NON-STRUCTURAL -> land use + enforcement + drills

CONTENTS -> secure fixtures equipment utilities

RULE -> combine; do not substitute one for all

ASCII MASTER FLOW – PANEL 7/12: Code-to-compliance rail

- 1 STANDARD / BYE-LAW
- 2 COMPETENT DESIGN
- 3 TRAINED CONSTRUCTION
- 4 APPROVAL -> INSPECTION -> MAINTENANCE

ASCII MASTER FLOW – PANEL 8/12: Ductility concept box

DEFORMATION CAPACITY

- > controlled energy dissipation
- > supports life-safety objective

BOUNDARY -> concept only, no detailing instruction

ASCII MASTER FLOW – PANEL 9/12: Retrofit priority funnel

SCREEN EXISTING STOCK

PRIORITISE LIFELINES + HIGH OCCUPANCY + HIGH VULNERABILITY

DETAILED ASSESSMENT -> SANCTION -> WORK -> VERIFY

AUDIT / LISTING ALONE -> NOT COMPLETION

ASCII MASTER FLOW – PANEL 10/12: Lifeline continuity map

HOSPITAL + EOC + FIRE / RESCUE

POWER + WATER + TELECOM

ROAD / BRIDGE / ACCESS

TEST -> physical safety AND post-event function

MUST REMEMBER: Earthquake risk combines seismic hazard, exposed people and assets,...

MUST REMEMBER: Earthquake risk combines seismic hazard, exposed people and assets,...

ASCII MASTER FLOW – PANEL 11/12: Monitoring-prediction boundary

SENSOR -> EVENT DETECTION -> PARAMETER REPORT

HAZARD ASSESSMENT -> PLANNING INPUT

PREPAREDNESS -> PROTECTIVE ACTION

NOT -> precise future magnitude place and time

CLOSE DISTINCTION: Magnitude is not intensity; earthquake occurrence is not disaster...

CLOSE DISTINCTION: Magnitude is not intensity; earthquake occurrence is not disaster...

ASCII MASTER FLOW – PANEL 12/12: Earthquake answer spine

DEFINE RISK -> SEPARATE MAGNITUDE INTENSITY

MAP ZONE + SITE + EXPOSURE + VULNERABILITY

COMBINE CODE RETROFIT LAND USE CONTENTS AND LIFELINES

TEST ENFORCEMENT EQUITY CONTINUITY AND VERIFIED OUTCOME

EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Source every magnitude, intensity, zone,...

EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Source every magnitude, intensity, zone,...